Advanced Empirical Finance: Topics and Data Science

Shrinkage for portfolio choice

Stefan Voigt

University of Copenhagen and Danish Finance Institute (DFI)

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Where we are

- Lecture 3: plug-in portfolios lose out of sample. On the Dow Jones stocks, 1/N beat the plug-in minimum-variance portfolio, and the plug-in tangency portfolio was not investable
- The estimates $\hat{\mu}$ and $\hat{\Sigma}$ are noisy, but they are not useless

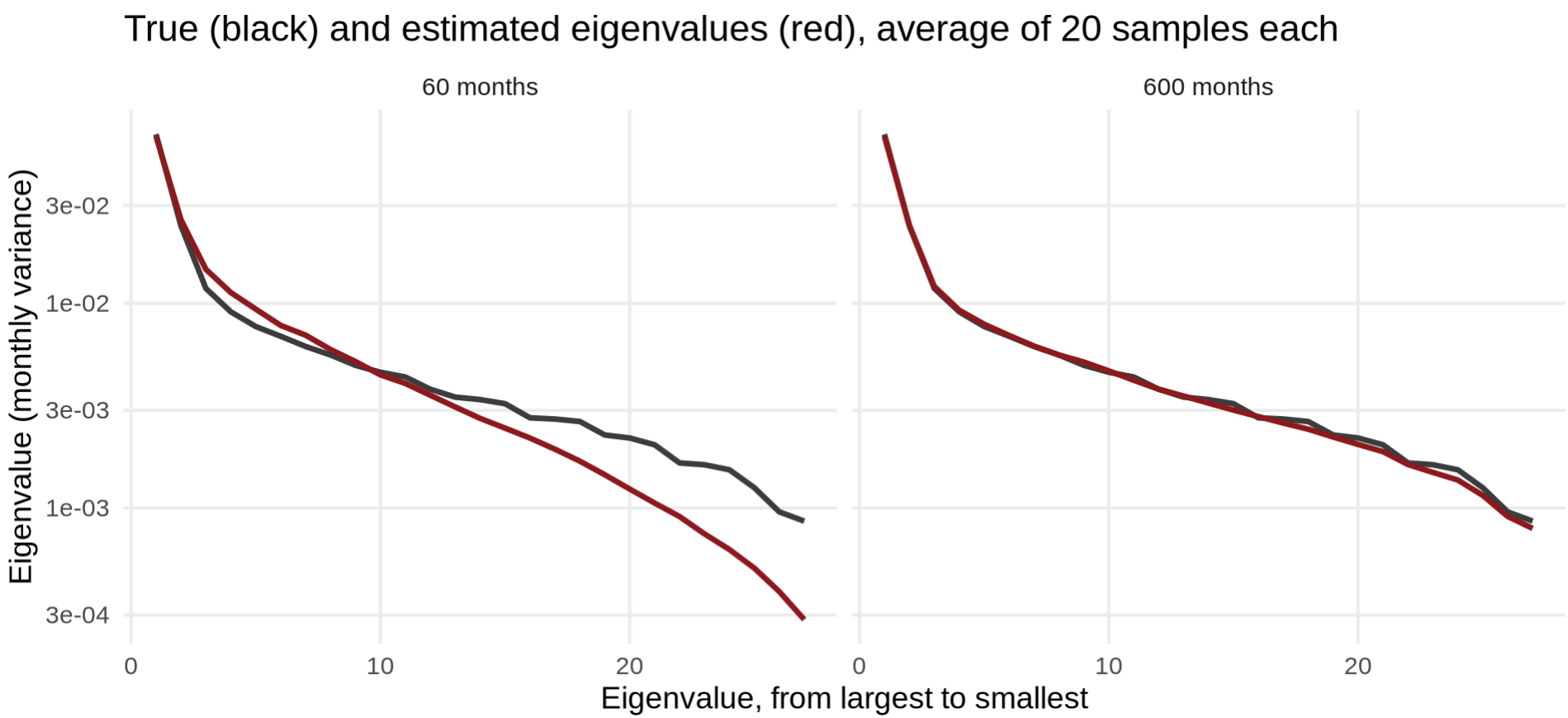
> **Today:** instead of throwing the estimates away, pull them toward something simpler. That is **shrinkage**: a little bias for much less variance

The covariance matrix in high dimensions

Too many parameters

- $\boldsymbol{\Sigma}$ has $N(N+1)/2$ parameters: 27 stocks give 378, estimated from 60 months of 27 returns
- With $N \geq T$, $\hat{\boldsymbol{\Sigma}}$ is singular and has no inverse
- With $N < T$, it is invertible but ill-conditioned: its **smallest eigenvalues are too small** and its largest too large
- $\hat{\boldsymbol{\Sigma}}^{-1}$ puts the most weight on the directions with the smallest eigenvalues: the portfolios that look almost riskless in the sample. Those are exactly the ones we estimate worst

Estimated eigenvalues are too spread out



Linear shrinkage

Whiteboard: shrinking toward a target

Combine the sample covariance matrix $\hat{\Sigma}$ with a simple target F:

$$\hat{\Sigma}_\delta = \delta F + (1-\delta)\hat{\Sigma}, \qquad \delta \in [0,1]$$

Which δ minimizes the expected squared error $E\,\|\hat{\Sigma}_\delta - \Sigma\|^2 = E\sum_{i,j}(\hat{\sigma}_{\delta,ij} - \sigma_{ij})^2$?

For an unbiased $\hat{\Sigma}$ and a fixed target:

$$\delta^* = \frac{E\,\|\hat{\Sigma} - \Sigma\|^2}{E\,\|\hat{\Sigma} - F\|^2} = \frac{\sum_{i,j}\mathrm{Var}(\hat{\sigma}_{ij})}{\sum_{i,j}E(\hat{\sigma}_{ij} - f_{ij})^2}$$

- More noise in $\hat{\Sigma}$: shrink more. A target far from the data: shrink less
- Ledoit and Wolf estimate both parts from the data, so $\hat{\delta}$ needs no tuning

Targets and the Ledoit-Wolf estimator

- **A multiple of the identity**, $F = mI$ with m the average variance (Ledoit & Wolf, 2004a): all variances equal, all correlations zero
- **Constant correlation**, the same correlation for every pair (Ledoit & Wolf, 2004b): the estimator of exercise 3
- **A single-factor model**, covariances from market betas (Ledoit & Wolf, 2003)

R

```r
# Ledoit-Wolf (2004a): shrink toward the average variance times the identity
ledoit_wolf <- function(x) {
  t <- nrow(x)
  n <- ncol(x)
  x <- sweep(x, 2, colMeans(x))
  sample <- crossprod(x) / t
  m <- sum(diag(sample)) / n
  d2 <- sum((sample - m * diag(n))^2) / n
  b2 <- sum(map_dbl(seq_len(t), \(i) sum((tcrossprod(x[i, ]) - sample)^2))) / (n * t^2)
  delta <- min(b2, d2) / d2
  list(sigma = delta * m * diag(n) + (1 - delta) * sample, delta = delta)
}
```

Targets and the Ledoit-Wolf estimator

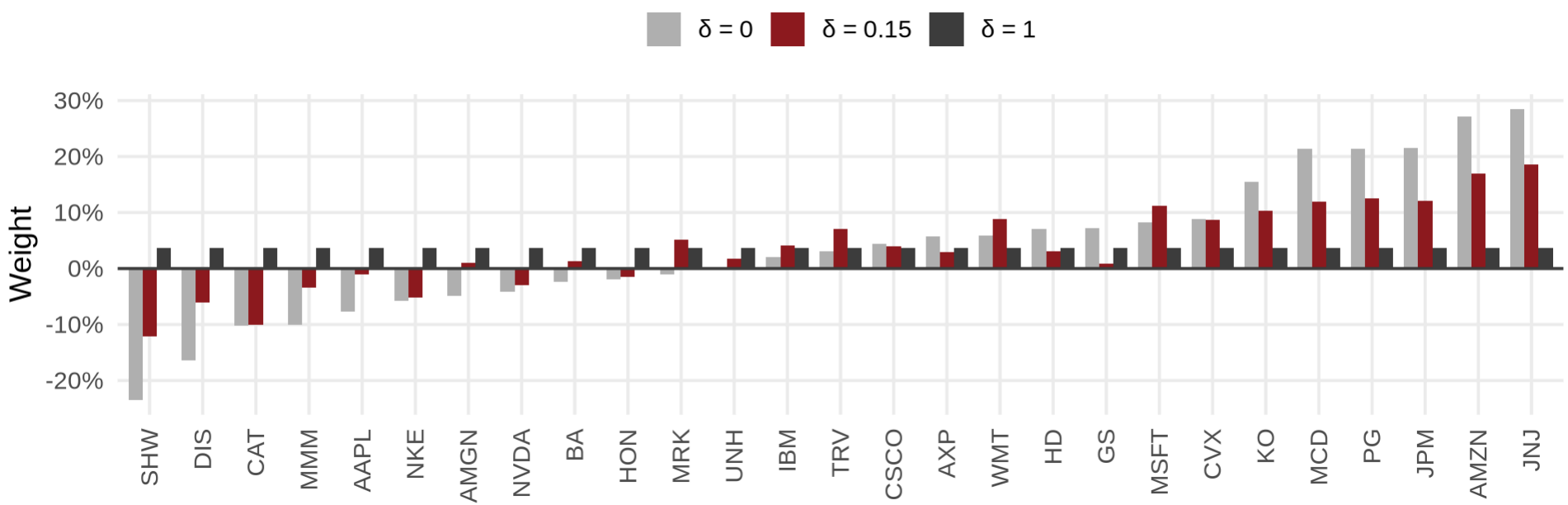
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Python

```
# Ledoit-Wolf (2004a): shrink toward the average variance times the identity
lw = LedoitWolf().fit(window)
sigma_lw = lw.covariance_
delta = lw.shrinkage_
```

Shrinkage moves the minimum-variance portfolio

- With the target mI, the minimum-variance portfolio of the target is **1/N**: shrinkage moves the weights from the plug-in portfolio ($\delta = 0$) toward 1/N ($\delta = 1$)
- Ledoit-Wolf for the latest window: $\hat{\delta}$ = 0.15



Constraints as shrinkage

Whiteboard: why no short sales help (Jagannathan & Ma, 2003)

Minimum variance with $\omega_i \geq 0$. With multipliers $\lambda_i \geq 0$, and $\lambda_i = 0$ whenever $\omega_i > 0$, the first-order conditions are

$$\Sigma\omega - \lambda = \lambda_0 \iota$$

- $\omega_{\text{no short}}$ is the **unconstrained** minimum-variance portfolio of

$$\tilde{\Sigma} = \Sigma - (\lambda\iota' + \iota\lambda')$$

- For a stock the optimizer wants to short, the constraint lowers its variance by $2\lambda_i$ and its covariances by $\lambda_i + \lambda_j$: the constraint shrinks the large covariances that would have made the weight negative
- Wrong constraints, in theory, but in practice they shrink much like a covariance target does

Shrinking expected returns

Whiteboard: Bayes-Stein (Jorion, 1986)

Prior: all expected returns are equal, to the mean of the minimum-variance portfolio μ_{mvp}. How far should $\hat{\mu}$ move toward $\mu_{\text{mvp}}\,\iota$?

$$\hat{\mu}_{\text{BS}} = (1-w)\,\hat{\mu} + w\,\mu_{\text{mvp}}\,\iota, \qquad w = \frac{N+2}{(N+2) + T\,(\hat{\mu}-\mu_{\text{mvp}}\iota)' S^{-1} (\hat{\mu}-\mu_{\text{mvp}}\iota)}$$

with $S = \frac{T-1}{T-N-2}\hat{\Sigma}$

- The more dispersed the means look relative to their noise, the less we shrink
- **Empirical Bayes:** the data choose the center and the strength of the prior

Bayes-Stein in code

R

```r
# Bayes-Stein (Jorion, 1986): shrink toward the mean of the mvp
bayes_stein <- function(x) {
  t <- nrow(x)
  n <- ncol(x)
  mu <- colMeans(x)
  s_inv <- solve(cov(x) * (t - 1) / (t - n - 2))
  mu_mvp <- sum(s_inv %*% mu) / sum(s_inv)
  d <- mu - mu_mvp
  w <- (n + 2) / ((n + 2) + t * as.numeric(t(d) %*% s_inv %*% d))
  list(mu = (1 - w) * mu + w * mu_mvp, w = w)
}
```

For the latest 60-month window of the Dow Jones stocks: w = 0.53. Bayes-Stein moves the expected returns about halfway to their common mean

Bayes-Stein in code

Python

```python
# Bayes-Stein (Jorion, 1986): shrink toward the mean of the mvp
def bayes_stein(x):
    t, n = x.shape
    mu = x.mean(axis=0)
    s_inv = np.linalg.inv(np.cov(x, rowvar=False) * (t - 1) / (t - n - 2))
    mu_mvp = (s_inv @ mu).sum() / s_inv.sum()
    d = mu - mu_mvp
    w = (n + 2) / ((n + 2) + t * d @ s_inv @ d)
    return (1 - w) * mu + w * mu_mvp, w
```

For the latest 60-month window of the Dow Jones stocks: w = 0.53. Bayes-Stein moves the expected returns about halfway to their common mean

The Bayesian view

- Bayes-Stein is **empirical Bayes**: the prior's center and strength come from the data
- A full Bayesian investor treats μ and Σ as unknown and averages over them: the **predictive distribution** of returns
- With a diffuse prior, returns look riskier than $\hat{\Sigma}$ says, so the investor holds less in risky assets (Klein & Bawa, 1976)
- Priors can also come from economics, e.g. from an asset pricing model. An overview: Avramov & Zhou (2010)

Back to the Dow Jones

Five strategies

- **1/N**, the **plug-in minimum-variance** portfolio and the **minimum-variance portfolio without short sales** from lecture 3
- **Ledoit-Wolf minimum variance:** the minimum-variance portfolio of $\hat{\Sigma}_{\text{LW}}$
- **Bayes-Stein and Ledoit-Wolf tangency:** the tangency portfolio of $\hat{\mu}_{\text{BS}}$ and $\hat{\Sigma}_{\text{LW}}$

R

```r
mvp <- function(sigma) {
  weights <- solve(sigma, rep(1, ncol(sigma)))
  weights / sum(weights)
}

tangency <- function(mu, sigma) {
  weights <- solve(sigma, mu)
  weights / sum(weights)
}
```

Five strategies

- **1/N**, the **plug-in minimum-variance** portfolio and the **minimum-variance portfolio without short sales** from lecture 3
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Python

```python
def mvp(sigma):
    weights = np.linalg.solve(sigma, np.ones(sigma.shape[0]))
    return weights / weights.sum()

def tangency(mu, sigma):
    weights = np.linalg.solve(sigma, mu)
    return weights / weights.sum()
```

The backtest

R

```
backtest <- map(seq(window_length + 1, nrow(excess_matrix)), \(t) {
  window <- excess_matrix[(t - window_length):(t - 1), ]
  sigma_hat <- cov(window)
  sigma_lw <- ledoit_wolf(window)$sigma
  returns_next <- excess_matrix[t, ]
  weights <- list(
    "1/N" = rep(1 / n_stocks, n_stocks),
    "Plug-in minimum variance" = mvp(sigma_hat),
    "Ledoit-Wolf minimum variance" = mvp(sigma_lw),
    "Minimum variance, no short sales" = min_variance_no_short(sigma_hat),
    "Bayes-Stein and Ledoit-Wolf tangency" = tangency(bayes_stein(window)$mu, sigma_lw)
  )
  tibble(
    date = excess_wide$date[t], strategy = names(weights), weights = weights,
    ret_excess = map_dbl(weights, \(w) sum(w * returns_next))
  )
}) |>
  bind_rows()
```

The loop is the one from lecture 3; `excess_matrix`, `min_variance_no_short()` and the other inputs as there

The backtest

Python

```
records = []
for t in range(window_length, n_months):
    window = excess_matrix[t - window_length:t]
    sigma_hat = np.cov(window, rowvar=False)
    sigma_lw = LedoitWolf().fit(window).covariance_
    weights = {
        "1/N": iota / n_stocks,
        "Plug-in minimum variance": mvp(sigma_hat),
        "Ledoit-Wolf minimum variance": mvp(sigma_lw),
        "Minimum variance, no short sales": min_variance_no_short(sigma_hat),
        "Bayes-Stein and Ledoit-Wolf tangency": tangency(bayes_stein(window)[0], sigma_lw),
    }
    for strategy, w in weights.items():
        records.append({"date": excess_wide["date"][t], "strategy": strategy,
                        "ret_excess": w @ excess_matrix[t]})
backtest = pl.DataFrame(records)
```

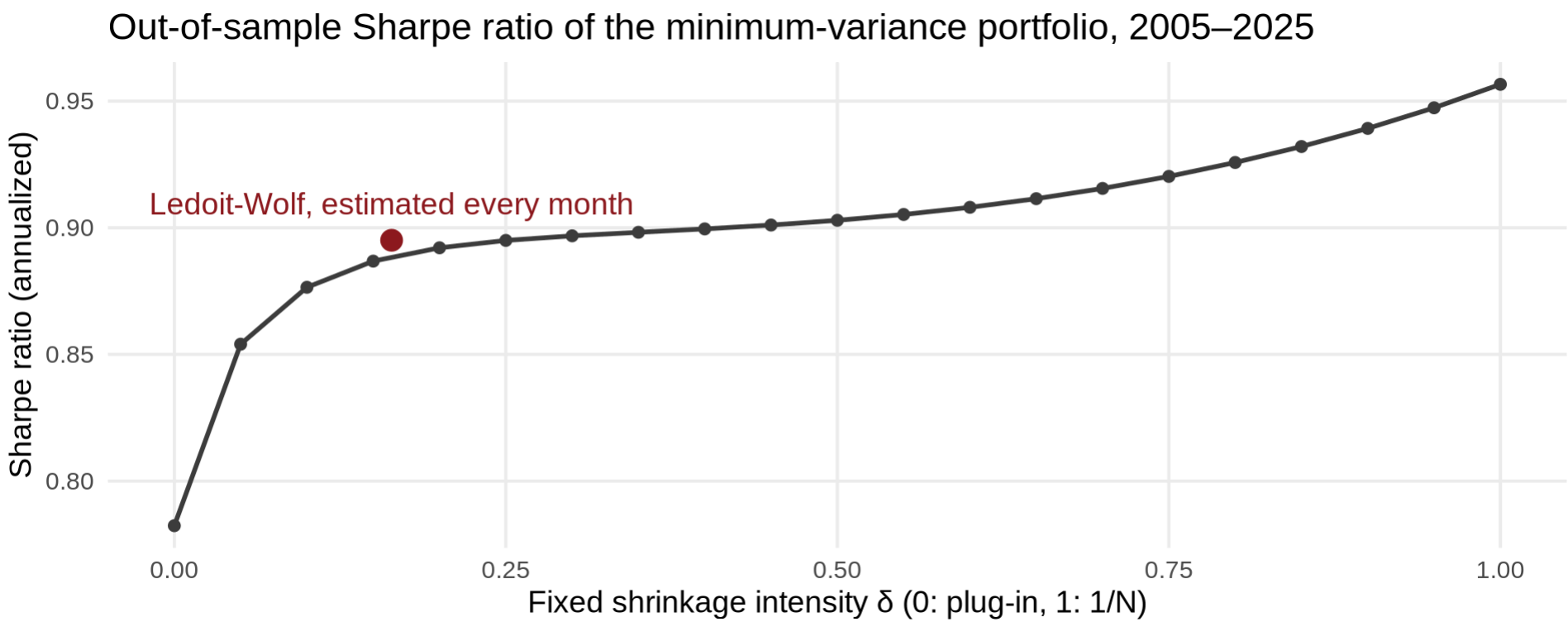
The loop is the one from lecture 3; `excess_matrix`, `min_variance_no_short()` and the other inputs as there

Out of sample, February 2005 to December 2025

Strategy	Excess return	Volatility	Sharpe ratio	Turnover per month
1/N	14.3%	14.9%	0.96	4%
Ledoit-Wolf minimum variance	11.5%	12.9%	0.90	23%
Minimum variance, no short sales	10.7%	12.4%	0.86	15%
Plug-in minimum variance	12.4%	15.8%	0.78	60%
Bayes-Stein and Ledoit-Wolf tangency	6.2%	59.3%	0.10	223%

- Annualized, excess returns over the T-bill rate. Ledoit-Wolf beats the plug-in and the no-short-sale minimum-variance portfolios. Shrinking the means helps the tangency portfolio, but not enough

How much shrinkage?



- Ledoit-Wolf chooses $\hat{\delta}$ between 0.10 and 0.23: optimal for estimating Σ, not for the portfolio. Here, more shrinkage pays all the way to 1/N, helped by the survivorship of today's Dow Jones stocks
- Shrinkage aimed at the portfolio's variance: Ledoit & Wolf (2017)

Shrinkage everywhere

- Shrinkage trades a little bias for much less variance. The same idea returns as **ridge regression and the lasso** in the machine learning lectures
- Constraints, priors and targets are all ways to tell the optimizer what we believe before we see the data
- **Next lecture:** transaction costs act like shrinkage, too, and parametric portfolio policies

> Assignment 1 on portfolio choice starts on 25 February